

Network Troubleshooting, Management, Monitoring, and Optimization

1. Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

ANS :

```
C:\Users\HARSH>ipconfig /all

Windows IP Configuration

Host Name . . . . . : HARSH-SIDHPURIYA
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Local Area Connection* 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC #4
Physical Address. . . . . : 52-8B-85-AC-6E-7C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 5:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC #5
Physical Address. . . . . : 56-0B-05-AC-6E-7C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet1
Physical Address. . . . . : 00-50-56-C0-00-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::54ba:4a02:fe34:ed65%13(Preferred)
IPv4 Address. . . . . : 192.168.247.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 25 June 2026 13:53:51
Lease Expires . . . . . : 25 June 2026 17:03:12
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.247.254
DHCPv6 IAID . . . . . : 838881366
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-0A-3D-24-F8-ED-FC-F0-F9-20
NetBIOS over Tcpip. . . . . : Enabled
```

```
Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8
Physical Address. . . . . : 00-50-56-C0-00-08
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::afab:7e61:830d:9cb9%4(Preferred)
IPv4 Address. . . . . : 192.168.101.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 25 June 2026 13:53:51
Lease Expires . . . . . : 25 June 2026 17:03:12
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.101.254
DHCPv6 IAID . . . . . : 855658582
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-0A-3D-24-F8-ED-FC-F0-F9-20
Primary WINS Server . . . . . : 192.168.101.2
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC
Physical Address. . . . . : 50-0B-05-AC-6E-7C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::387c:745d:bc9c:cd0%21(Preferred)
IPv4 Address. . . . . : 192.168.0.101(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 25 June 2026 13:54:02
Lease Expires . . . . . : 25 June 2026 17:54:15
Default Gateway . . . . . : 192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DHCPv6 IAID . . . . . : 290503605
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-0A-3D-24-F8-ED-FC-F0-F9-20
DNS Servers . . . . . : 192.168.0.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek Gaming GbE Family Controller
Physical Address. . . . . : F8-ED-FC-F0-F9-20
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

➤ Windows IP Configuration

Parameter	Example Value
IPv4 Address	192.168.0.101
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server 1	192.168.0.1
DNS Server 2	8.8.4.4

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

ANS :

- When Wifi/Ethernet Disconnected

```
C:\Users\HARSH>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.
```

- After Reconnecting

```
C:\Users\HARSH>ping www.google.com

Pinging www.google.com [142.251.157.119] with 32 bytes of data:
Reply from 142.251.157.119: bytes=32 time=13ms TTL=117
Reply from 142.251.157.119: bytes=32 time=13ms TTL=117
Reply from 142.251.157.119: bytes=32 time=18ms TTL=117
Reply from 142.251.157.119: bytes=32 time=14ms TTL=117

Ping statistics for 142.251.157.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 18ms, Average = 14ms
```

- Short Note

- **When disconnected**, the ping command failed because the computer had no network connection or could not reach the DNS server.
- **After reconnecting**, the ping command received replies, confirming that the internet connection and network communication were working correctly.

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

ANS :

➤ **Steps**

1. Open **Task Manager** (Ctrl + Shift + Esc).
2. Go to **Performance** → **Wi-Fi/Ethernet**.
3. Play a YouTube video for 5 minutes.
4. Observe network usage.

➤ **Observation**

Item	Example
Duration	5 Minutes
Peak Network Usage	3.6 Mbps
Highest Bandwidth Process	Google Chrome (YouTube)

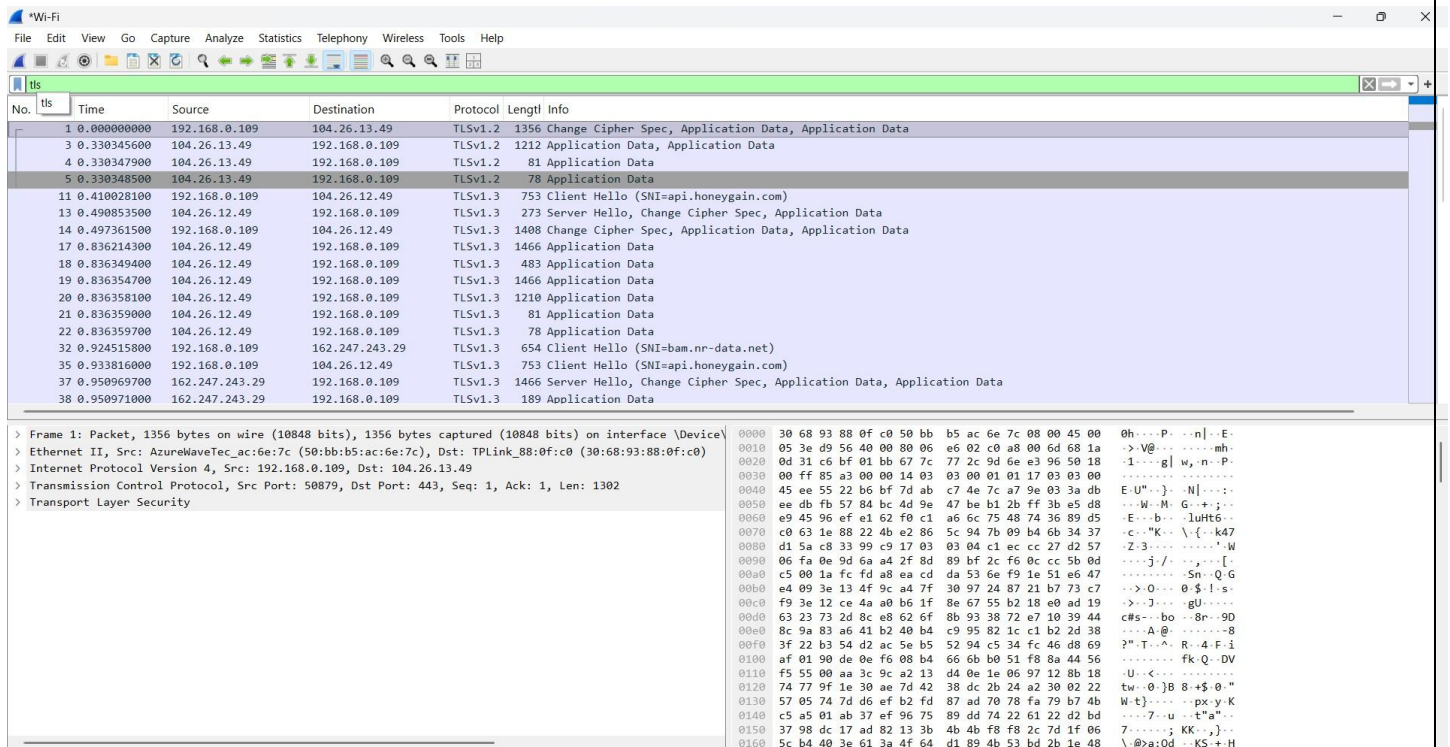


4. Download and install Wireshark
(or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets. Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.

ANS :

➤ **Steps**

1. Install **Wireshark**.
2. Start capturing on your active Wi-Fi/Ethernet interface.
3. Browse Flipkart or Instagram Web for 30 seconds.
4. Stop the capture.
5. Use the filter: **TLS**



➤ **Three Types of Information Visible**

Information	Description
Source and Destination IP Addresses	Shows where packets originate and where they are sent.
Protocol	Displays whether the traffic uses HTTP, HTTPS (TLS), TCP, etc.
Packet Size and Time	Shows packet length and timestamps for analyzing network performance.

5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

ANS :

➤ **Home Network Optimization Checklist**

1. Restart the router and modem.
2. Move the router to a central, open location.
3. Use a wired Ethernet connection for gaming whenever possible.
4. Disconnect unused devices from the Wi-Fi network.
5. Update the router's firmware regularly.
6. Use the 5 GHz Wi-Fi band if supported.
7. Close background applications or downloads that consume bandwidth.